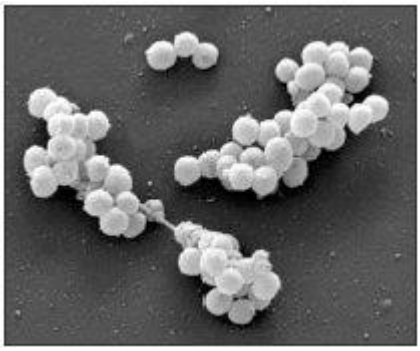
Q1. The photograph below shows bacteria that have developed resistance to antibiotics. They are called MRSA bacteria.



(a) When MRSA bacteria reproduce, they pass on their resistance to antibiotics to the next generation.

What part of a cell passes on information?

.....

1 mark

(b) MRSA bacteria can cause serious infections in people who are ill in hospital. The bacteria can live on a healthy person's **skin** or in their **lungs** without causing any harm.

Use this information to fill in the table below.
Suggest **two** ways MRSA bacteria can be spread from person to person.
Suggest how the spread of the bacteria can be prevented for each method.

	method of spread	**method of prevention**
1		
2		

2 marks

(c) People can be vaccinated against some diseases caused by bacteria or viruses.

Describe how vaccination prevents a person getting a disease.

.....

.....

.....

.....

.....

3 marks
maximum 6 marks

Mark schemes

Q1.

(a) • genetic material **or** genes **or** DNA

accept 'nucleus'
accept 'chromosomes'

1 (L7)

each method of prevention must be related to the method of spread for a mark
the method of spread **and** prevention are required for each mark
the same prevention method cannot be used for both methods of spread

(b) **method**

• touch **or** contact

prevention

any **one** from

• wash hands before **or** after touching a patient **or** before **or** after touching surfaces **or** articles

accept 'quarantine patients suspected of carrying disease'

• use alcohol rub when entering **or** leaving ward **or** room **or** hospital

accept 'change gloves between patients'
*accept 'stop contact with others' **or** 'limit visitors'*
*'cleaning' **or** 'washing hands' are insufficient*
'wearing gloves' is insufficient

1 (L7)

method

• coughing **or** sneezing

*accept 'through the air' **or** 'breathing on someone'*

prevention

• cover mouth **or** nose **or** use a handkerchief **or** wear a mask disease'

accept 'quarantine patients suspected of carrying
*accept 'stop contact with others' **or** 'limit visitors'*

1 (L7)

(c) any **three** from

• vaccine contains a dead **or** weakened form of the bacterium **or** virus

*accept 'inject with protein from the bacterium **or** virus'*
*'it contains a small amount of bacteria **or** virus' is insufficient*

• antibodies **or** antitoxins (are produced)

• (made by) white blood cells

accept 'white cells'

• further infections are prevented **or** bacteria **or** viruses **or** toxins are destroyed

*accept 'kills the bacteria **or** viruses'*
'stop you getting it again' is insufficient
'fights bacteria' is insufficient

'you become immune to the disease' is insufficient as it is given in the question

3 (L7)

[6]